Topic: Dividing Algebraic Expressions

EQ: How do I simplify algebraic expressions with division?

{Fun Fact} The division sign is another way of writing fractions. The dots on the top and bottom is where the numbers go $3 \div 4 = \frac{3}{4}$

Question	Working Out	Explanation
$8b \div 2$	① $\frac{8b}{2}$ ② $\frac{\cancel{8}b}{\cancel{2}}$ $8 \div 2 = 4$ ③ $b = b$ ④ $\boxed{4b}$	① Put the question into fraction form if it isn't already ② Divide the numbers or simplify the fraction ③ Cancel out any like variables (there is only b) ④ Put it together
$6d \div 4d$	① $\frac{6d}{4d}$ ② $\frac{6}{4} = \frac{3}{2}$ ③ $\frac{3\cancel{d}}{2\cancel{d}} = 1$ ④ $\frac{3}{2} \times 1 = \boxed{\frac{3}{2}}$	① same ↑ ② simplified the fraction (divided both by 2) ③ $\frac{d}{d}$ if there is a variable in the numerator and denominator we can cancel them out ∴ anything divided by itself is one ④ combine ($\frac{3}{2} \times 1$) is the same as $\frac{3}{2}$
$18ay \div 4a$	① $\frac{18ay}{4a}$ ② $\frac{18}{4} = \frac{9}{2}$ ③ $\frac{9\cancel{a}y}{2\cancel{a}}$ ④ $\boxed{\frac{9y}{2}}$	① same ↑ ② simplify the fraction (divided both by 2) ③ canceled out the a's y has no like terms in the denominator so nothing changes ④ combine

$\frac{50a^2}{5a}$	1 $\frac{50}{5} = 10$	1 already in fraction form! So we divide the numbers first
	2 $\frac{10a^{\cancel{2}}}{\cancel{a}} \to \frac{10 \times a \times \cancel{a}}{\cancel{a}}$	2 $a^2 = a \times a$ so we can cancel out <u>one</u> of the top a's with the bottom
	3 $\boxed{10a}$	

Summary:

To simplify algebraic equations with division we have to ______________________.

If there are two letters the same on top and bottom we ______________________.